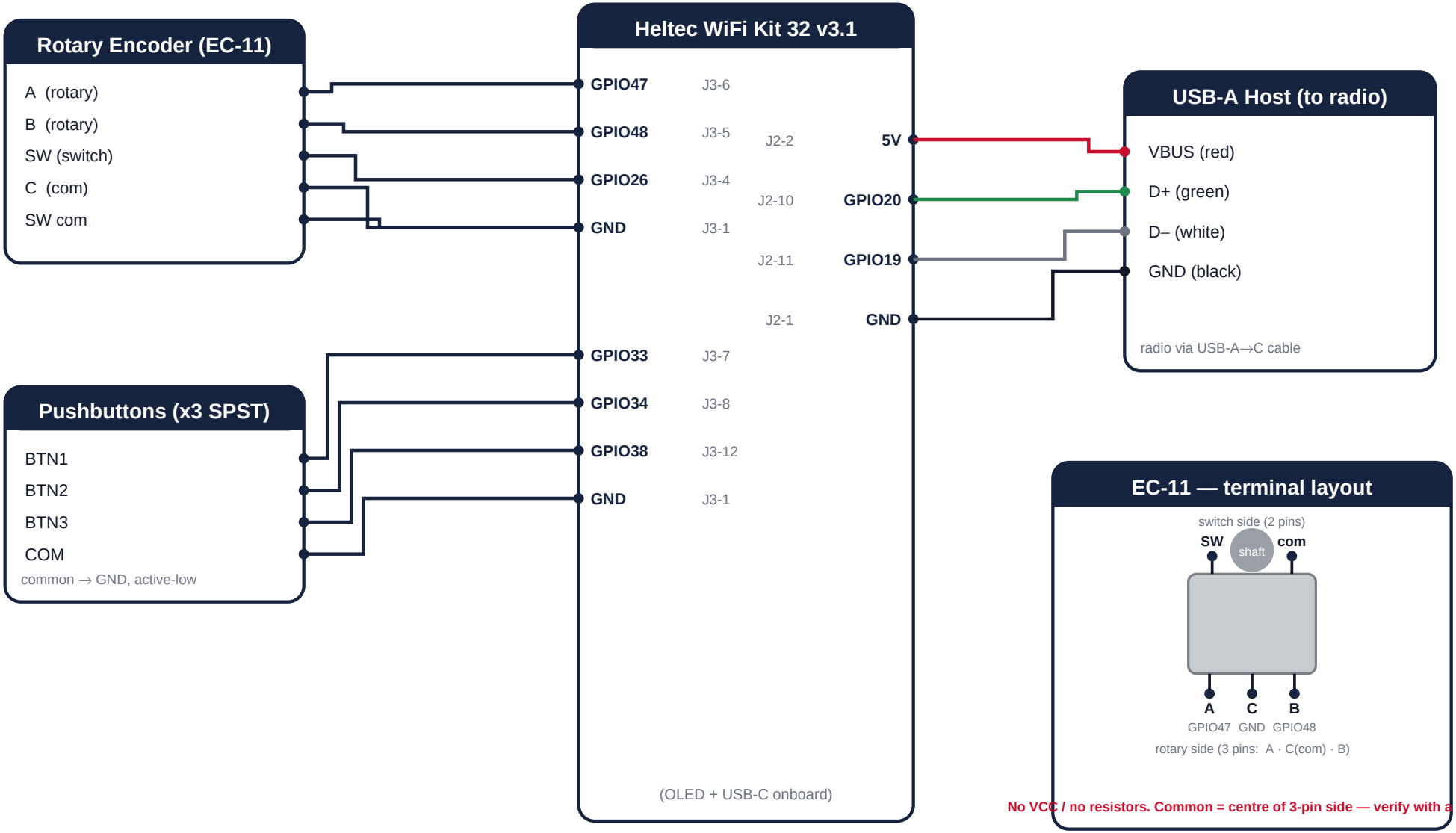
CW Bridge — POC Control Wiring

Heltec WiFi Kit 32 v3.1 (ESP32-S3) · EC-11 encoder + 3 buttons + USB-A host · header pins as J2-n / J3-n



EC-11 + all 3 buttons live on header J3; USB-A host on J2. EC-11 has no VCC — both encoder commons and the button COM all tie to J3-1.
A/B/SW use ESP32-S3 internal pull-ups (no resistors). Optional: 10 nF from A & B to GND tames bounce. USB D–/D+ fixed to GPIO19/20 — keep short & twisted.

CW Bridge — POC Wiring Tables & Function Map

Pin-for-pin connections (with header pins) and firmware control map

EC-11 rotary encoder → header J3

EC-11 pin	GPIO	Pin	Note
A (rotary)	GPIO47	J3-6	channel A (CLK)
B (rotary)	GPIO48	J3-5	channel B (DT)
SW (switch)	GPIO26	J3-4	push switch
C (common)	GND	J3-1	rotary common
SW common	GND	J3-1	switch common

Pushbuttons (3x SPST) → header J3

Button	GPIO	Pin	Function
BTN1	GPIO33	J3-7	CW memory 1 (CQ)
BTN2	GPIO34	J3-8	CW memory 2 (TU 73)
BTN3	GPIO38	J3-12	Health overlay (long)
COM	GND	J3-1	common (shared)

Notes

- EC-11 is a bare encoder: no VCC pin, no onboard resistors.
- Rotary common (C, centre of 3-pin side) + switch leg + button COM → J3-1.
- A/B/SW use ESP32-S3 internal pull-ups — no external resistors.
- Optional: 10 nF from A & B to GND smooths contact bounce.
- EC-11 clones vary — confirm the common pin with a meter.
- Keep USB D–/D+ short & twisted (full-speed 12 Mbit/s).
- Do NOT wire GPIO 8–14 (reserved on LoRa-equipped boards).

USB-A host → radio → header J2

USB-A pin	GPIO	Pin	Note
VBUS (red)	5V	J2-2	bus power to radio
D+ (green)	GPIO20	J2-10	USB D+ (fixed)
D– (white)	GPIO19	J2-11	USB D– (fixed)
GND (black)	GND	J2-1	ground

Firmware control map (POC)

Control	Action	Detail
Encoder press	cycle	WPM ↔ AF gain
Encoder rotate	adjust	active target (KS / AG0nnn;)
Encoder long	menu	open on-screen menu
BTN1 / BTN2	play	CW memory slot 1 / 2 (KY)
BTN1/2 long	abort	stop in-flight CW

! Memory buttons TRANSMIT

BTN1/BTN2 key the radio's CW keyer (KY) and transmit on the current freq/mode. Use a dummy load, set CW mode, be clear before testing. Long-press the button to abort.